FlightBERT: Binary Encoding Representation for Flight Trajectory Prediction

Dongyue Guo, Edmond Q. Wu, *Senior Member, IEEE*, Yuankai Wu, *Member, IEEE*, Jianwei Zhang, Rob Law, and Yi Lin, *Member, IEEE*

***Abstract*— Flight Trajectory Prediction (TP) is an essential task in Air Traffic Control (ATC). Currently, the TP task is usually achieved by regression approaches, which concatenates several scalar attributes of the observation into a low-dimensional vector as the inputs. However, it is difficult to accurately model aircraft motion patterns using low-dimensional features in complex and time-varying ATC environments. To improve the performance of the TP task, in this paper, a novel framework, called FlightBERT, is proposed based on Binary Encoding (BE) representation, which enables us to tackle the TP task as a multi binary classification problem. Specifically, the scalar attributes of the flight trajectory are encoded into binary codes and transformed into a high-dimensional representation by the attribute embedding module. Considering the prior knowledge among flight attributes, an Attribute Correlation Attention (ACoAtt) block is designed to explicitly capture the correlations among the specific attributes. A stacked Transformer block is applied to serve as the backbone network, which is followed by the predictor to generate the outputs. Considering the nature of flight trajectory, a hybrid constrained loss, i.e., combining the mean square error loss with the binary cross-entropy loss, is innovatively designed to optimize the proposed framework. The proposed method is validated on a large-scale dataset, which is collected from the real-world ATC environment. The experimental results demonstrate that the proposed method outperforms other baselines by quantitative and qualitative evaluations.**

***Index Terms*— Attribute correlation attention, binary encoding representation, flight trajectory prediction, multi binary classification, transformer.**

Manuscript received 6 February 2022; revised 23 June 2022 and 28 August 2022; accepted 1 November 2022. Date of publication 10 November 2022; date of current version 8 February 2023. This work was supported in part by the National Natural Science Foundation of China (NSFC) under Grant U20A20161, Grant 62001315, and Grant U2241228; in part by the Open Fund of Key Laboratory of Flight Techniques and Flight Safety, Civil Aviation Administration of China (CAAC), under Grant FZ2021KF04; and in part by the Fundamental Research Funds for the Central Universities under Grant 2021SCU12050. The Associate Editor for this article was M. Gao. *(Corresponding author: Yi Lin.)*

Dongyue Guo is with the National Key Laboratory of Fundamental Science on Synthetic Vision, Sichuan University, Chengdu 610000, China (e-mail: dongyueguo@stu.scu.edu.cn).

Edmond Q. Wu is with the Key Laboratory of System Control and Information Processing, Ministry of Education of China, the Shanghai Engineering Research Center of Intelligent Control and Management, and the Department of Automation, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: edmondqwu@163.com).

Yuankai Wu is with the Department of Civil Engineering, McGill University, Montreal, QC H3A 0C3, Canada (e-mail: yuankai.wu@mcgill.ca).

Jianwei Zhang and Yi Lin are with the College of Computer Science, Sichuan University, Chengdu 610000, China (e-mail: zhangjianwei@scu.edu.cn; yilin@scu.edu.cn).

Rob Law is with the Asia-Pacific Academy of Economics and Management, University of Macau, Taipa, Macau 999078, China (e-mail: roblaw@um.edu.mo).

Digital Object Identifier 10.1109/TITS.2022.3219923

I. Introduction

FLIGHT Trajectory Prediction (TP) is one of the fundamental research in the Air Traffic Control (ATC) procedure. It is a crucial technique of various applications in the ATC domain, such as air traffic flow prediction [1], conflict detection [2], [3], arrival time estimation [4]. Accurate TP is vital to help the Air Traffic Controller (ATCo) to achieve real-time airspace management in a more efficient manner, which further ensures the safety of air traffic operations. Most importantly, after the Trajectory-Based Operation (TBO) conception was proposed by the Single European Sky ATM Research (SESAR) and New Generation Air Transportation Systems (NextGen), the TP technique also serves as a powerful way to enhance free-flight operation, thus attracting increasing research attention from all over the world [5].

TP refers to the estimation of flight status (mainly includes longitude, latitude, altitude, velocity, etc.) at the future time instant. In general, capturing influential transition patterns from flight trajectories is the key to achieving the TP task, which is learned to predict future flight positions. According to the time scale, the flight TP task can be divided into three categories, i.e., long-term, medium-term, and short-term [6]. For long-term TP, the flight trajectory (or route) is determined by the flight plan with other required information, as well as the strategic measures. For medium-term TP, weather information and real-time traffic situation make primary impacts on the trajectory prediction. For short-term TP, the current motion status of the aircraft is the decisive factor to the future flight trajectory, where spatial and temporal patterns can be mined from historical data.

In this work, we mainly focus on the short-term TP task which is the key technique to ensure the safety of air traffic operations [7]. Great efforts have been made to achieve the TP task, and existing TP methods can be summarized into the following categories: physics-based, model-driven, and data-driven methods. The physics-based methods build a series of kinematic equations to model the flight patterns based on aerodynamics, in which the aircraft is regarded as a mass point [8], [9]. However, these methods typically rely on the aircraft performance collected from the ideal test environment, which cannot be accurately measured for real-time air traffic systems and thus limits its actual performance. Similarly, the model-driven approaches were well-designed based on the prior knowledge of experts, which shows poor adaptability and generalizability facing unseen environments. The

1558-0016 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

representative model-driven methods in the TP task are based on the Kalman Filter (KF) and its variants [10], [11], which estimate the next state according to the current observation. In the past decade, thanks to the rapid development of deep learning, the neural architecture based data-driven methods are promising to learn the implicit knowledge by training with a large amount of historical trajectory data and have achieved state-of-the-art performance [6], [12], [13], [14].

Traditionally, the TP task is defined as a sequence regression problem. In practice, the scalar attributes of the flight status (such as longitude, latitude, altitude, and time) are concatenated to a low-dimensional vector as the trajectory representations and serve as the model inputs, and the detailed attributes of the future flight trajectory are also predicted in scalars. However, current solutions failed to address the following issues to achieve a high-confidence TP task:

1) All the representations of flight trajectory are denoted by scalars with different units (i.e., degree for longitudinal, meter for altitude, etc.), which cannot provide discriminative features among inputs with respect to time instants, and thereby limits the capabilities of the data-driven approaches to mine high-level abstract patterns.
2) For existing approaches, normalization algorithms are widely applied to unify the contributions of input dimensions on the final outputs. However, the normalized prediction errors are required to be transformed into raw data range to measure the actual performance, which shows a huge vulnerability impacted by normalization algorithms and data range. Specifically, the invalid inputs will occur and make failed predictions when the observed data is out of the predefined data range of the normalization algorithms. In addition, the normalization algorithm will reduce the data discrimination, especially in the case of a large region of interest (ROI), and the prediction error will deteriorate in the inverse transformation process.
3) The explicit correlations between the specific attributes of the trajectory are also not fully considered to utilize data specificities. For example, the offsets (latitude, longitude, and altitude) between flight positions are positively correlated with the velocity in the corresponding dimensions.
4) As to the ATC work, it is the multi-step TP that can make practical contributions to real-time ATC decisions [7]. However, since the vulnerability of the predicted results tends to lead to accumulating errors, it is difficult to implement the multi-step TP task by the regression-based TP approach.

To enhance the TP performance and further support the air traffic operation, in this work, a novel deep learning based framework, called Binary Encoding Represented Flight Trajectory prediction (FlightBERT), is proposed to address the aforementioned issues properly.

Firstly, we propose a Binary Encoding (BE) representation to encode the scalar values of the flight trajectory attributes. The BE-based trajectory can enlarge the scalar representation into a higher dimension, which is also able to provide discriminative features and improve corresponding information capacity. In this way, the scalar attributes (i.e., longitude, latitude, altitude, and the corresponding velocity vectors) of the flight trajectory are encoded to a set of binary codes, which further can be regarded as an *n-hot* vector. It is clear that the transformation between the BE representation and the actual flight attributes is unique and can be achieved in a straightforward way.

In succession, to focus on different attributes of input trajectory, independent attribute embedding blocks are designed to capture abstract high-level knowledge, and also formulate unified attribute embedding vectors with the same dimension. To utilize the prior knowledge of the correlations among specific attributes of flight trajectory, a novel attention block, called Attribute Correlation Attention (ACoAtt), is proposed to explicitly capture the influences of velocity attribute on the corresponding positional dimension. A Transformer-based backbone network is improved to capture spatial and temporal transition patterns, which are further applied to predict the future flight positions by a predictor network.

In this proposed framework, both the input and output trajectory are organized in the BE representations. Therefore, the TP task is innovatively defined as a Multi Binary Classification (MBC) problem, i.e., predicting the binary class for each dimension of the output *n-hot* vectors. During the model optimization, the Binary Cross-Entropy (BCE) loss is naturally applied to improve the classification accuracy for the MBC task. Meanwhile, considering the final performance is measured in scalar, the Mean Squared Error (MSE) loss is also introduced to measure the prediction errors by transforming the BE representation to the scalar attributes. Hence, a Hybrid Constrained (HC) loss function is proposed to optimize the model parameters by evaluating the prediction errors for both the BE representations and the decimal values.

To validate the proposed framework, extensive experiments are conducted on a large-scale trajectory dataset collected from the real-world ATC environment. The experimental results demonstrate that the proposed framework outperforms other baseline models. Furthermore, the BE representation has good generalizability, i.e., not only performed well with Transformer architectures, but also achieved expected performance with LSTM networks. Meanwhile, the ACoAtt block is also validated on different model architecture to support our motivation of utilizing prior knowledge.

In summary, this paper contributes the flight trajectory prediction in the following ways:

- A novel TP framework, called FlightBERT, is proposed to achieve the TP task, in which a BE representation is proposed to formulate the TP task as a multi binary classification problem. To the best of our knowledge, this is the first work to tackle the TP tasks by using a series of binary classification tasks.
- The proposed BE representation is benefits to mining high-dimensional trajectory features, which also can be generalized to other model architectures. In addition, compared to the normalization-based approaches, the proposed BE representation is robust to the data range

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

since the transformation between BE and scalar is a uniquely determined process.
- An ACoAtt block is dedicatedly proposed to consider the prior knowledge among the data specificities of the flight trajectory, which is important to mine attribute correlations to improve the TP performance.
- A BCE and MSE combined loss is innovatively proposed to consider both the classification accuracy and the final performance evaluation of the MBC-based TP task.
- Extensive experiments are conducted on a real-world dataset to validate the proposed approach, including all technical improvements. Most importantly, the proposed BE representation provides insights into the TP task in future studies.

The remainder of this paper is organized as follows. The related works of flight trajectory prediction and deep learning are reviewed in Section II. The proposed FlightBERT framework is detailly described in Section III. The experiments, results, visualizations, and discussions are presented in Section IV. The conclusion and future work are described in Section V.

II. Related Work

A. Flight Trajectory Prediction

1) Physics-Based Methods: The physics-based methods typically obtained the aircraft performance parameters from the existing database and make some aerodynamics and kinematics assumptions based on the current aircraft status. One of the most famous is the Base of Aircraft Data (BADA) database [15] which provides the Aircraft Performance Model (APM) and Operating Procedure Coefficients (OPCs) of various aircraft types. The Trajectory Computation Infrastructure (TCI) capabilities and interfaces were presented in [16], which is based on the APM of the BADA. The proposed TCI computed the trajectory by considering aircraft intent, earth model, and APMs. In [17], the OPCs were represented as a set of discrete variables and established a three-dimensional point mass model to emulate the aircraft dynamics. However, the BADA is not always suitable for all cases, some aircraft performance parameters are not publicly available due to business sensitivity. Towards this end, an aircraft mass estimation method was studied in [18]. The authors model the aircraft mass by the kinetic model at the takeoff moment and the motion of the aircraft on the runway. In [8], the concept of velocity profile was introduced to enhance the dynamic model for the TP task in the aircraft taxiing stage.

Although the aircraft performance models are able to be estimated through algorithms, systematic errors also will be introduced inevitably due to real-time traffic dynamics. It is clear that the physics-based methods still have many limitations in practical applications without an accurate aircraft performance model.

2) Model-Driven Methods: The KF-based algorithm is one of the widely used model-driven methods in the early TP tasks. The KF can achieve the optimal estimation by giving system equations, and real-time observations. In [19], the KF algorithm was introduced to tackle the flight state estimation problem and simulated based on the Automatic Dependent Surveillance-Broadcast (ADS-B) data. The Unscented Kalman Filter (UKF) algorithm was applied to achieve the flying target tracking [10]. To solve the restrictions of the single model estimation in the complex ATC environment, the Interacting Multiple Model (IMM) was proposed and has widely been used in the field of target tracking [20].

Although the model-driven methods have been performed successfully in the field of target tracking, they are sensitive to the noise of observations, and the cumulative error will be multiplied in the multi-step prediction process.

3) Data-Driven Methods: The data-driven methods are expected to learn the transition patterns of the flight motion from a large number of historical flight trajectory data, which can be classified into three categories, i.e., data mining based, machine learning based, and deep learning based. In [21], the typical trajectory library was abstracted from historical radar data by leveraging the data mining algorithms to improve the TP performance for ATC. The Hidden Markov Model (HMM), the typical machine learning method, was used to mine the flight trajectory patterns in [22]. A switched dynamical HMM was proposed to solve the trajectory classification task in [23]. In addition, the Gaussian Mixture Model [24], Gaussian Processes [25], Support Vector Machine [26], and some variants algorithms were also introduced into the TP tasks and demonstrated that effectiveness by experiment.

With significant progress achieved in other applications by deep learning approaches, such as Computer Vision (CV) [27], [28], natural language processing (NLP) [29], automatic speech recognition (ASR) [30], [31], enormous deep learning based TP studies have been emerged in the past decade [13], [14], [32], [33]. In [12], the backpropagation (BP) neural network was adopted to solve the high-precision four-dimensional TP task. To model the uncertainty in the ATC environments, the Bayesian neural network was introduced to consider the environmental factor in [34] and [35]. A hybrid architecture was proposed to extract the spatial-temporal features of the trajectory data by combining Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) network [36]. However, these methods usually only model on the specific airspace, such as airport [37], terminal airspace [4], [13], or single route [36]. A constrained LSTM Network (cLSTM) was proposed in [6], which considers climbing, cruising, and descending/approaching phases constraints. In addition, the flight plan and weather are also incorporated into the neural network to enhance the performance of the TP tasks [38], [39].

B. Deep Learning

Deep learning based TP solutions are typically formulated as time-series prediction problems. It is well known that the Recurrent Neural Networks (RNNs) and its variants have shown significant performance for sequential data modeling in a variety of tasks, such as text classification [40], ASR [41]. To solve the gradient vanishing and exploding problems of the RNNs in learning with long-term dependencies, the LSTM network is designed to capture the long-term and short-term dependencies by the proposed memory cells [42]. Currently,

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

the LSTM layer has become a standard component in the deep learning based TP task [6], [14], [32].

The Transformer was first proposed for machine translation tasks in 2017 [43]. Compared with the LSTM, the transformer computes the representations entirely using the self-attention mechanism and modeling the sequential features by the position embedding. Due to amazing performance obtained by Transformer architecture in the NLP domain, it was also introduced into other fields (such as CV [28], ASR [31]) and achieved expected performance. Here, we introduce the Transformer architecture as the backbone network for the proposed framework which is expected to capture the long-term dependencies from the historical trajectory data.

Binary classification is a common task in the field of machine/deep learning [44]. The core idea of optimizing the *n-hot* vector outputs through multi binary classification is also inspired by the multi-label learning works [45].

III. Methodology

A. Problem Formulation

In this work, we focus on the short-term flight trajectory prediction task, i.e., forecasting the aircraft state in the subsequent a few time steps by the given flight trajectory observation sequence. Let a trajectory observation sequence $P_{1:T-1} = \{p_1, p_2, \ldots, p_{t-1}\}$, the objective of our model is to generate $Y_{T:T+N} = \{y_t, y_{t+1}, \ldots, y_{t+n}\}$. The p_t, y_t represents the observed and predicted trajectory point of the time step t, respectively, which are a vector composed of a number of scalar trajectory attributes. The N is the number of prediction steps in the inference process.

Specifically, in this work, a flight trajectory point p_t is defined as follows:

$$p_t \triangleq [Lon_t, Lat_t, Alt_t, Vx_t, Vy_t, Vz_t] \tag{1}$$

where the $Lon_t, Lat_t, Alt_t, Vx_t, Vy_t, Vz_t$ corresponds to longitude, latitude, altitude, and velocity in x, y, z dimensions of the trajectory point, respectively. The x, y, z dimensions of the velocity corresponds to the longitudinal, latitudinal, and altitudinal attributes, respectively.

Further, the multi-step prediction task can be refined as the Eq. (2) and Eq. (3), where the $\mathcal{F}$ is the TP model. In other words, the TP model inputs the observation sequence of time step 1 to $t-1$ and outputs the predicted aircraft state y_t. The predicted y_t will append into the input sequence of the TP model which further serves as the observation trajectory (pseudo) in the y_{t+1} inference procedure.

$$y_t = \mathcal{F}(P_{1:T-1}) \tag{2}$$

$$y_{t+n} = \mathcal{F}(\{P_{1:T-1}, \hat{Y}_{T:T+N-1}\}),$$
$$\hat{Y}_{T:T+N-1} = \{y_t, \ldots, y_{t+n-1}\} \tag{3}$$

B. The Proposed FlightBERT Framework

Existing deep learning based flight TP tasks typically be formulated as regression problems. As discussed in Section I, there are some challenges in the regression based TP tasks. In this work, we propose the BE representation to transform the trajectory attributes into a set of discrete representations, i.e., *n-hot* vectors. Further, based on the BE representation, a novel framework, called FlightBERT, is proposed to tackle the TP task by the MBC-based approach.

The architecture of the proposed FlightBERT framework is illustrated in Fig. 1, consisting of the binary encoding/decoding process, Flight Trajectory Prediction Network (FTPNet), and Hybrid Constrained Loss (HC Loss). Specifically, the trajectory points $P_{1:T-1}$ are firstly encoded into BE representations. In succession, the proposed FTPNet projects the inputs into a compacted space for trajectory embeddings and learns the high-level trajectory representations to generate the outputs. The independent trajectory embedding module is designed to convert the *n-hot* vectors of the trajectory point to compacted embeddings. We propose to apply the Transformer as the backbone network that is followed by a predictor network. The predicted BE representations of trajectory point to further transform to scalar attributes by a binary decoding process. Finally, the prediction errors between the prediction y_t and the ground truth p_t are calculated by the HC loss function, which are backpropagation to hidden layers for optimizing the FTPNet parameters.

C. Binary Encoding Representation

A common representation for the trajectory point is a four-dimensional (4D) vector, including longitude, latitude, altitude, and time, which can be obtained by concatenating the scalar attributes of the observation. For the 4D vector representations, it is difficult to learn the implicit semantic features among the attributes of the trajectory point, as well as the sequential points from low-dimensional representation for a neural network. In practice, the transition of the aircraft velocities are expected to be estimated from the historical observations implicitly, which is intractable to current approaches. Fortunately, the velocity is available in the surveillance data and is an important factor affecting the position of the aircraft in future time steps. Existing approaches failed to leverage the velocity to facilitate the TP task due to unique units for different attributes.

To this end, the following observed attributes, including longitude, latitude, altitude, and velocity in the corresponding dimensions obtained from the surveillance data, are regarded as the inputs of the FlightBERT framework (Eq. (1)). In addition to the positional attributes, we also predict the velocity vectors to enhance the applicability of the proposed framework. A novel BE representation is innovatively proposed to represent the attributes of the trajectory point, which is easily projected to the high-dimensional feature space. The BE representation is able to eliminate the impacts of different units and also enhance the information capability for data mining tasks.

In general, the proposed BE representation is to convert each attribute from a decimal number to a binary vector (with only 0 and 1). The BE process can be described as follows. Firstly, all decimal attributes are rounded to a certain precision by multiplying 10^m (m is the number of decimals to be retained), with the goal of transforming the float attributes to integers. Secondly, the transformed attributes are encoded into a set

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

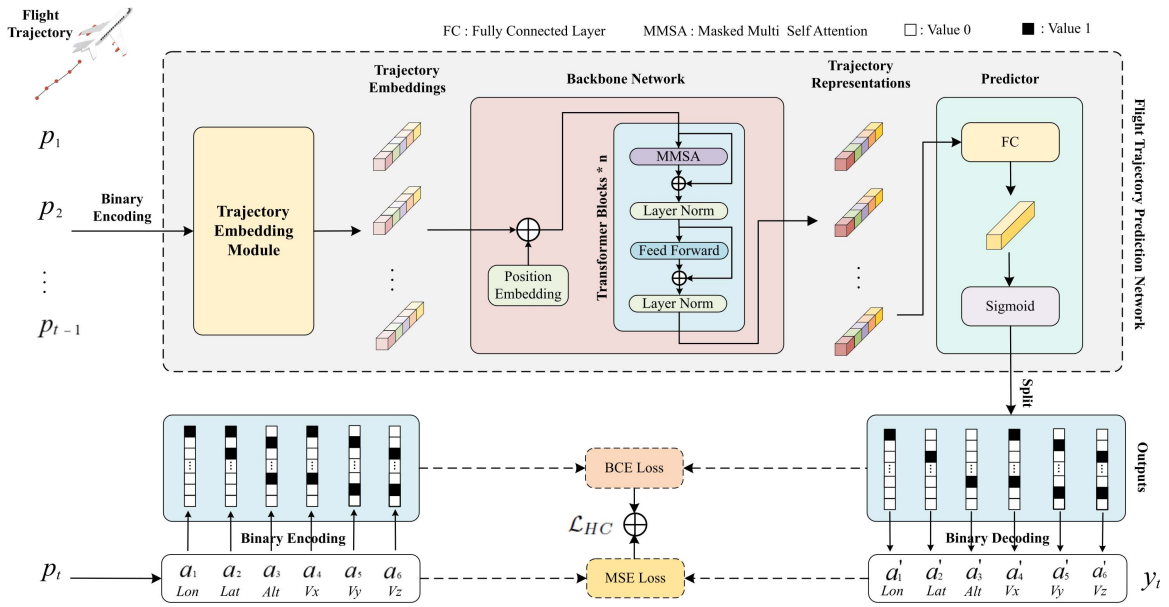


Fig. 1. Proposed FlightBERT framework.

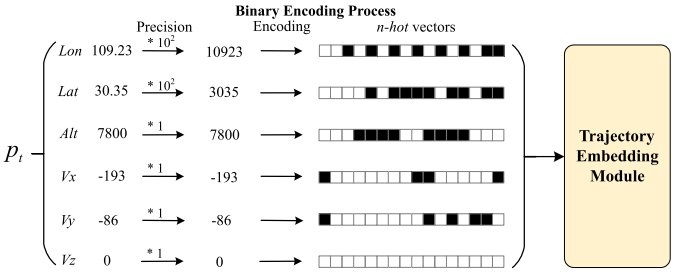


Fig. 2. Example of the BE process.

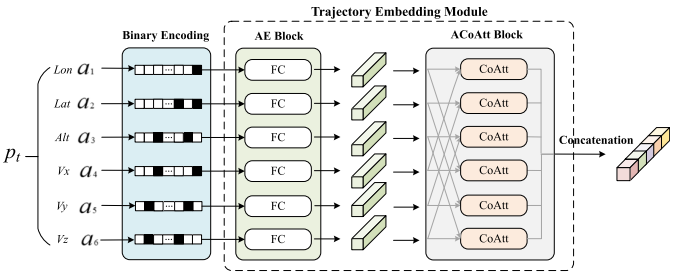


Fig. 3. Architecture of the trajectory embedding module.

of binary codes by the base-2 numeral system. Note that the velocity attributes are the signed integers to denote the flight headings, and the highest bit of the BE representation is used to denote the sign of the numbers. In this work, the highest bit is 1 if the value of the velocity is a negative number, while it is set to 0 if the value is a positive number.

For instance, let a trajectory point p_t is [109.23, 30.35, 7800, -193, -86, 0], the transformation process of the proposed BE representation can be illustrated in Fig. 2. In order to facilitate the understanding of the proposed BE representation, a 16-bit vector is applied to denote each attribute of the trajectory point. It is clear that the number of bits for each attribute can be specifically defined by considering the range of attribute items and their required precision for certain tasks. It is noted that the proposed method also can be adapted to different input precision (i.e., the values of m) by re-setting the number of encoding bits.

Obviously, the BE representation of each attribute can be seen as an *n-hot* vector and the p_t is a joint *n-hot* vector that is generated by concatenating BE representation of all attributes. Thus, the proposed BE representation is able to convert the scalar values to the vector representation without any reduction of precision. The BE representation is denoted by a high-dimensional vector to facilitate the capability of model learning for neural architectures.

D. Flight Trajectory Prediction Network

As shown in Fig. 1, the proposed FTPNet is constructed by using three modules, i.e., trajectory embedding module, backbone network, and a predictor network.

1) Trajectory Embedding Module: Although the proposed binary encoding representation transforms the scalar attributes to the *n-hot* vector representation, it is still a challenging task to learn the discriminative features and the correlations from these intra and inter *n-hot* vectors. Specifically, the intra *n-hot* vectors represent different trajectory attributes whereas the inter *n-hot* vectors represent different trajectory points. An intuitive way is to project the BE representations into the embedding space, with compacted and abstract features.

To this end, independent trajectory embedding modules are designed to project the BE representations of each attribute into the embedding space and capture correlations among the attributes. Fig. 3 illustrates the architecture of the proposed trajectory embedding module which consists of Attribute Embedding (AE) block and Attribute Correlation Attention (ACoAtt) block. Specifically, several FC layers are used to project the trajectory attributes to the high-dimensional representations in the proposed AE block. The ACoAtt block are designed to capture the inherent correlation between the specific attributes by the proposed Correlation Attention (CoAtt) mechanism.

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

Based on the prior knowledge of the kinematic patterns between two continuous time instant, we can see that the longitude, latitude, altitude of the next time step are mainly positively correlated to the velocity in the x, y, z dimensions at current time instant. Similarly, the velocity in the x, y, z dimensions are also correlated with the positional sequence (longitude, latitude, altitude). It is believed that the correlations between the positional information and velocity not only are capable of improving the TP performance, but also enhancing the explainability of neural blocks based on the kinematic patterns.

In this work, an ACoAtt block is innovatively proposed to utilize the prior knowledge to enhance TP performance. The architectures of the ACoAtt block and CoAtt module are illustrated in Fig. 4. The CoAtt module is dedicated to recalibrating the attribute embedding to explicitly capture the correlations by inputting the representations of two specific-correlated attributes. The proposed CoAtt module is implemented by the multi-head attention mechanism and the concatenating operation.

Mathematically, let trajectory point $p_t = \{a_1, a_2, a_3, a_4, a_5, a_6\}$, and the a_k^b is the BE representation of the attribute k, respectively. The inference rules of trajectory embedding e_t can be described as follows. As shown in Eq. (4) the attribute embeddings e_{a_k} are outputted by the FC layers. And then the correlation embedding e'_{a_k} is computed by the multi-head attention mechanism as Eq. (5), where the Q is the linear projection sequence of the e_{a_k}, K, V are linear projection sequences of the correlated attribute embedding e_{a_j}. The final attribute embedding e''_{a_k} is a joint vector by concatenating the e_{a_k} and e'_{a_k} (Eq. (6)).

$$e_{a_k} = FC(a_k^b) \tag{4}$$

$$e'_{a_k} = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{5}$$

$$e''_{a_k} = concat(e_{a_k}, e'_{a_k}) \tag{6}$$

Note that different CoAtt modules are designed to recalibrate the embedding of each attribute, independently. The cross-correlations for correlated attributes (e.g, *Lon* and Vx) are achieved in a bidirectional manner.

Finally, all the final attribute embeddings e''_{a_k} are concatenated to formulate the trajectory point embedding e_t, which further be fed into the backbone network.

2) Backbone Network: In this work, an improved Transformer architecture [46] is proposed to serve as the backbone network for the trajectory sequence modeling which stacks multi-layer Transformer decoder block [43]. As shown in Fig. 1, every Transformer block shares the same architecture, constructing by a masked multi-head self-attention (MMSA) module, layer normalization, and position-wise feedforward layer. The reference architecture has achieved applause performance in many language modeling tasks. Similar to the language modeling in [46], the TP task in our work is also a sequence prediction problem by giving the input context of the trajectory representations. In our work, the main purpose to introduce the Transformer architecture is to learn the high-dimensional trajectory representations from the inputted trajectory embeddings.

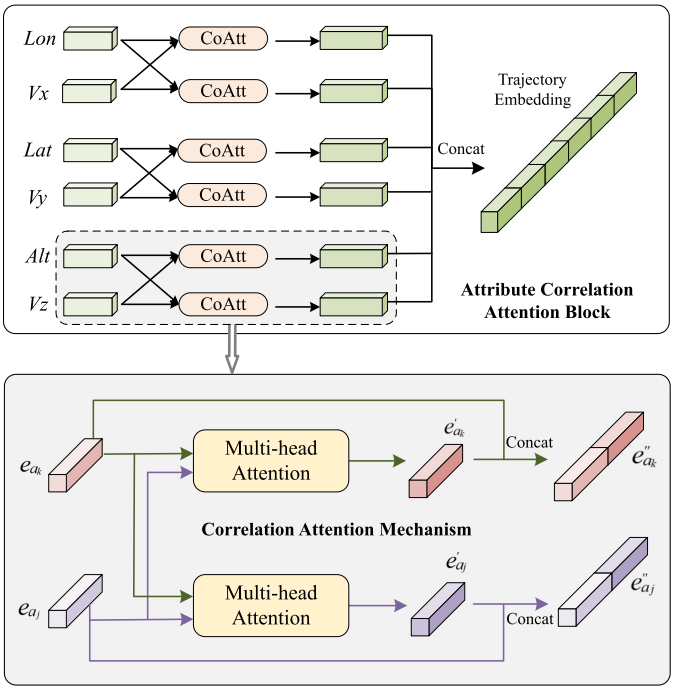


Fig. 4. Proposed attribute correlation attention block and correlation attention mechanism.

In practice, the RNN based network, such as LSTM, and BiLSTM, also are competitive alternatives of the backbone network. However, the Transformer provides a more effective architecture to cope with long-term dependencies through the self-attention mechanism. It can be dynamically generated the weights (capture semantic correlations) according to the pairwise similarity among the input sequence. Compared with Transformer blocks, the RNN based architectures perform predictions in the autoregressive manner which are known to suffer from fatal error accumulation [47]. In this work, to evaluate the generalization of the proposed framework, we also discuss other alternatives of the backbone network in Section IV.

3) Predictor Network: As shown in Fig. 1, the predictor network is an FC layer with the Sigmoid activation. The FC layer is used to project the hidden representations extracted from the historical trajectory sequence into a prediction vector, with the dimension being the sum of the BE representation bits of the required trajectory attributes.

The outputs of the FC layer are activated by the Sigmoid function and further split into several specific vectors according to the corresponding trajectory attribute dimensions.

In the binary decoding process, these specific vectors are quantized from continuous space into discrete space (BE representations) by a threshold function. Specifically, let a vector $V = \{v_0, v_1, \ldots, v_i\}$ split from the model outputs and V_b is an initial BE representation for V. The V_{b_i} will be set to 1 if $v_i > 0.5$, otherwise set to 0. Finally, the binarized BE representations are decoded into the real values (decimals), which are further used to evaluate the training or prediction errors.

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

E. Hybrid Constrained Loss

Benefiting from the proposed BE representation, we formulate the TP task as a number of the binary classification problems. The BCE loss is intuitively applied to minimize the errors of the BE representation between the prediction and the ground truth. However, the BE representation decomposes the model outputs as a number of independent binary classification tasks, which ignores the intra and inter correlations of the attributes for a certain flight position.

As an MBC-based task, the binary misclassifications of the BE representation provide identical impacts on the final BCE loss, which contributes to distinct prediction errors for the final TP performance. Specifically, the errors on the high bit of the BE representation denote larger decimal errors, while the lower bit is only for smaller errors. *For instance*, a 16-bit BE representation for the altitude (measured by meters), the misclassification of 1st bit denotes only 1 meter, while the misclassification of 10th bit denotes up to 1024 meters (i.e., 2^{10}).

To address the mentioned issues, an intuitive way is to simultaneously consider the classification accuracy and corresponding impacts on decimal errors. By referring to the common regression tasks, the MSE loss is also combined with the BCE loss to formulate the hybrid constrained loss for the model optimization. Let the p, y and p^b, y^b are the real-values (decimals) and the BE representations of the ground truth and prediction, respectively. The proposed HC loss can be mathematically described as Eq. (7) - Eq. (11):

$$\mathcal{L}_{HC} = \mathcal{L}_{MSE}(p, y) + \mathcal{L}_{BCE}(p^b, y^b) \tag{7}$$

$$\mathcal{L}_{MSE}(p, y) = \sum_{i=1}^{K} \mathcal{L}_{MSE}^{a_k}(p_{a_k}, y_{a_k}) \tag{8}$$

$$\mathcal{L}_{MSE}^{a_k}(p_{a_k}, y_{a_k}) = \frac{1}{N}\sum_{n=1}^{N}(p_{a_k} - y_{a_k})^2 \tag{9}$$

$$\mathcal{L}_{BCE}(p^b, y^b) = \frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} \mathcal{L}_{BCE}^{a_k}(p_{a_k}^b, y_{a_k}^b) \tag{10}$$

$$\mathcal{L}_{BCE}^{a_k}(p_{a_k}^b, y_{a_k}^b) = -\frac{1}{M}\sum_{i=1}^{M}(p_{a_k}^b[i] * log(y_{a_k}^b[i]) + (1 - p_{a_k}^b[i]) * log(1 - y_{a_k}^b[i]))) \tag{11}$$

where the N is the number of samples in the minibatch, K represents the number of trajectory attributes, i.e., $K = 6$ in this work. The p_{a_k}, y_{a_k} represents the decimal values of the kth attribute in the ground truth and prediction, respectively. Similarly, the $p_{a_k}^b$, $y_{a_k}^b$ are the BE representations, while M saves the dimension of the *n-hot* vector of each attribute.

The proposed HC loss constrains the outputs by the sum of the BCE loss and MSE loss. The BCE loss minimizes the classification error by considering the BE representation of the ground truth and prediction (binarized outputs), while the MSE loss constrains the decimal errors by considering real-value (decimals) between the ground truth and prediction. In this way, the proposed HC loss is expected to reduce the high bit prediction errors and enhance the intra and inter correlations of the BE represented trajectory attributes. By reducing the decimal errors, the proposed framework is expected to achieve higher performance for multi-step prediction since the prediction results will be regarded as pseudo observations.

IV. Experiment and Result

A. Dataset Preprocessing and Description

In this work, a real-world dataset is constructed to validate the proposed approach, which was collected from an industrial ATC system in China, from February 1 to March 15, 2021 (a total of 45 days). The following attributes of the flight trajectory are extracted to support the experiments in this work, including timestamps, call-sign, longitude, latitude, altitude, velocity in x, y, z dimensions. In the preprocessing, the call-sign is regarded as an identification to correlate the flight positions into a trajectory with 20 seconds intervals.

After the above preprocessing, a total of 144,605 flight trajectories are selected into our dataset to validate the proposed framework. The range of the interested longitude and latitude are [93.883, 113.689] and [19.305, 37.585], respectively. The flight position (longitude, latitude, and altitude) are in degree with the WGS-84 coordinate system, while the velocity in x, y, z dimensions are measured by Km/H with the Cartesian coordinates. The flight time of the trajectory in the dataset is from half an hour to two hours based on their real-time operations. The constructed dataset is split into train, validation, and test set by the following rules: the first 40 days are regarded as the training set, the 41st day is for validation and the rest 4 days are for the test set. The data splits are applied to all the experiments in this work to keep the experimental fairness.

B. Experimental Design and Configuration

1) Experimental Design: To demonstrate the effectiveness of the proposed FlightBERT framework, three group experiment are designed in our work. The experiments of group A are existing TP methods based on the deep regression framework and Kalman Filter. In this work, these methods severed as baselines, and the detailed description are listed as follows:

- A1: The LSTM-based regression networks are typically used in the TP task [6], [14]. In this experiment, the deep regression framework is constructed by stacking 8 layer LSTM network with 128 neurons. The inputs and the outputs are the six-dimensional (6D) vector concatenated by the six trajectory attributes. All the attributes are scaled to [0, 1] by the Max-Min normalization strategy.
- A2: The BiLSTM network is selected to alternate the LSTM layers in the baseline A1, and other configurations remain unchanged.
- A3: A Kalman Filter [48] based method is conducted to achieve the optimal state estimation of the flight trajectory using observed measurements, which is also a classical model-driven algorithm of the TP task.

To validate the proposed FlightBERT framework for precision generalizability, the experiments of the group B are designed as follows:

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

- B1: The resolution of the longitude and latitude is set to 0.001° (about 110 meters) and training with the proposed FlightBERT framework.
- B2: The resolution of the longitude and latitude is set to 0.01° (about 1100 meters) and other configurations are the same as B1.

In group C, to validate the robustness of the FlightBERT framework, the LSTM networks serve as the backbone network to replace the Transformer blocks, which achieves the TP task by the MBC-based approach.

- C1: The 8-layer LSTM network is applied to replace the Transformer blocks and serve as the backbone network in this experiment.
- C2: Similar to C1, the 8-layer BiLSTM network is alternatively selected to build the backbone network.

2) Experimental Configuration: In the experiment, the resolution and the number of bits of the BE representation of each trajectory attribute are listed as follows:

- Longitude and latitude: we use 18 and 16 bits to encode the real values (decimals) of longitude and latitude into BE representation for experiment B1, respectively, while 16 and 14 bits are used for experiment B2.
- Altitude: the altitude is measured in $10m$, and encoded into the binary codes with 11 bits.
- Velocity: the velocity is measured in Km/H, and encoded into the binary codes with 11 bits. Note that the highest bit indicates the sign of the real value.

With the above configurations, both the input and output of the proposed framework are with 78-dimensional vector (B1) or 74-dimensional vector (B2). In the proposed framework, all the attribute embedding e_{a_k} and correlation embedding e'_{a_k} sizes are set to 64. So, the size of trajectory embedding e_i is 768 which is the sum of the final attribute embeddings size. A 4-head multi-attention operator is applied in the ACoAtt block. A total of 8 stacked Transformer blocks are applied to build the backbone network. The output size of the FC layer in predictor is the sum of all the attribute BE representation bits, i.e. same with the input dimension.

In the training phase, the number of the input trajectory points is set to 9, i.e., using the lastest 3 minutes observations to predict the flight status of the next time steps. The Adam optimizer with 10^{-4} initial learning rate is applied to train the proposed FTPNet. Since the quantization process by the threshold function is a discontinuous operation in the binary decoding, the straight-through estimator (STE) [49] is applied to copy the gradients during the MSE loss backpropagation. In the multi-step inference process, a sliding window is applied to keep the input trajectory length. Based on the ATC work, we report the results with 1, 3, 9, 15 prediction time steps in Section IV-C, corresponding to 20 seconds, 1 minute, 3 minutes, and 5 minutes trajectory in the future.

In this work, we build the proposed FlightBERT framework and baselines with the open-source deep learning framework PyTorch 1.4.0. The models are trained on the server configured with Ubuntu 16.04 operating system, 4*NVIDIA GeForce RTX 2080 GPU, Intel(R) Core(TM) i7-7820X@3.6GHz CPU, and 32 GB memory.

3) Evaluation Metrics: In this work, the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are applied to evaluate the proposed methods and baselines, which are the common evaluation criteria of the TP tasks. The MAE intuitively reflects the absolute value of errors between ground truth and the prediction while the RMSE is more sensitive to the outliers predictions. Since some trajectory points are with altitude 0 or Vz 0, the Mean Absolute Percentage Error (MAPE) is not considered as an evaluation metric. The MAE and RMSE are defined as follows:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|a - a'| \tag{12}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(a-a')^2} \tag{13}$$

where the a, a' are the real-value (decimals) of trajectory attributes in the ground truth and prediction, respectively.

In addition, since the high real-time requirements of the TP task in ATC operations, the computational performance of the proposed methods and baselines are also evaluated in the experiments. The Average Time Costs (AvgTC) is computed as follows:

$$AvgTC = \frac{total_time_costs}{N * batch_size} \tag{14}$$

where N is the number of batches of the training or testing process.

C. Results and Quantitative Analysis

1) Experimental Results: The results of group A and group B are reported in Table I. In general, the proposed framework outperforms the baseline models for all the trajectory attributes in terms of both the MAE and RMSE. With the increase of the prediction steps, all comparative approaches suffer from performance reduction due to pseudo inputs. The proposed framework also achieves expected performance improvement for the multi-step prediction task. Meanwhile, higher precision for the BE representation is able to further enhance the TP performance for the proposed approach.

Specifically, for the deep regression-based baselines, thanks to the bidirectional sequence modeling, the performance of the BiLSTM network (A2) is superior to that of the LSTM network (A1) when the prediction step is 1 in both the MAE and RMSE metrics. However, the prediction errors of the BiLSTM network are larger than that of the LSTM network with the increase of the prediction steps. These results can be attributed that the BiLSTM model fails to complete bidirectional modeling by using pseudo trajectory points for the input sequence. The RNN based models are known to suffer from significant error accumulation effects due to its autoregressive inference. In the multi-step prediction process, the accumulated errors impose extra burdens to capture required transition patterns with pseudo inputs (like that of in real input data), and thereby impacting the model performance. The KF-based method (A3) works better than models A1 and A2 in 1-step prediction on MAE metrics but got worse performance when performing

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

TABLE I
RESULTS OF THE EXPERIMENT GROUP A AND B

Methods	Prediction steps	MAE ↓						RMSE ↓					
		Lon	Lat	Alt	Vx	Vy	Vz	Lon	Lat	Alt	Vx	Vy	Vz
A1 (LSTM / regression)	1	0.0164	0.0070	2.07	3.89	4.27	0.48	0.0194	0.0100	4.78	10.61	11.53	1.20
	3	0.0183	0.0087	3.99	7.08	7.85	0.73	0.1103	0.0979	10.15	20.02	21.23	1.68
	9	0.0290	0.0177	10.97	14.28	16.05	1.10	0.1997	0.1774	21.94	34.52	36.44	2.21
	15	0.0448	0.0306	18.31	19.38	22.00	1.27	0.2635	0.2321	34.02	43.13	45.56	2.46
A2 (BiLSTM / regression)	1	0.0118	0.0078	2.13	4.24	4.40	0.49	0.0147	0.0108	4.80	10.64	11.43	1.19
	3	0.0164	0.0109	4.00	7.96	8.08	0.82	0.1101	0.0984	10.09	20.14	21.21	1.73
	9	0.0342	0.0226	13.26	16.57	16.63	1.79	0.2015	0.1790	24.14	35.22	36.59	2.76
	15	0.0558	0.0374	27.49	23.19	23.07	2.42	0.2625	0.2301	43.58	44.85	46.18	3.35
A3 (Kalman Filter)	1	0.0091	0.0037	1.55	4.89	5.23	0.57	0.0687	0.0189	7.15	12.80	13.68	1.50
	3	0.0148	0.0066	3.13	9.23	9.81	0.99	0.1070	0.0301	11.35	23.71	24.97	2.47
	9	0.0356	0.0018	9.18	22.19	23.48	2.04	0.2307	0.0684	25.98	53.49	55.96	4.84
	15	0.0613	0.0335	16.35	34.18	36.43	2.92	0.3639	0.1115	41.78	81.10	85.29	7.05
B1 (Transformer / MBC / 0.001)	1	0.0039	0.0033	1.36	3.26	3.38	0.39	0.0558	0.0486	10.59	11.09	10.88	1.31
	3	0.0058	0.0051	2.52	5.14	5.39	0.55	0.0763	0.0696	13.79	15.98	16.50	1.73
	9	0.0125	0.0110	6.97	9.10	9.77	0.83	0.0985	0.0885	23.36	25.02	26.80	2.24
	15	0.0208	0.0187	11.95	11.98	12.96	0.95	0.1199	0.1067	34.02	30.48	32.80	2.43
B2 (Transformer / MBC / 0.01)	1	0.0042	0.0042	1.41	3.28	3.38	0.39	0.0553	0.0630	10.80	11.20	10.86	1.34
	3	0.0063	0.0057	2.58	5.11	5.39	0.56	0.0809	0.0769	14.12	16.05	16.53	1.76
	9	0.0140	0.0122	7.06	9.10	9.71	0.84	0.1187	0.1067	23.92	25.22	26.74	2.27
	15	0.0232	0.0207	12.00	12.08	12.97	0.95	0.1470	0.1282	34.45	30.87	32.86	2.45

the multi-step prediction due to losing the real measurements to update its parameters. In addition, the estimating accuracy of the KF-based model will decrease and converge slowly when the flight with great maneuvering characteristics, such as turning and climbing.

For the proposed framework (group B), as discussed in Section III-C, the proposed BE representation can be adapted to different input precision. As reported in the results, the performance is slightly decreased when the resolution of the longitude and latitude are set to 0.01°. Thanks to the BE representation, the proposed framework reduces the accumulated errors for each step, which further enhances the multi-step prediction for the TP task. As shown in Table I, even for a lower precision (0.01°), the proposed framework is still able to harvest better results of performing the multi-step prediction over the 1-step prediction in baselines for some attributes. However, we also recognize that the proposed framework achieves the TP task with larger RMSE metrics, which will be further discussed with the experiments of group C.

The experimental results of group C are illustrated in Table II. It can be seen that the proposed framework can also obtain desired performance improvement by incorporating the LSTM-based backbones. We can conclude that the proposed framework has remarkable generalization ability (including the BE representation and MBC mechanism), which can achieve desired performance improvement on both the Transformer and LSTM-based architectures. These results can ultimately confirm the effectiveness and efficacy of the proposed framework by decoupling the architecture dependencies. Considering the BE representation is the foundation of the MBC-based TP framework, we believe that the proposed BE representation can be generalized to various neural architectures and regarded as a robust representation to replace the scalar values in the TP tasks.

By using the BE representation, the LSTM-based models obtain comparable performance even over the Transformer architecture, as compared in C1 v.s. B1 and C2 v.s. B1, which indicates that the temporal dependencies of the flight trajectory should be considered for the TP task in a proper way.

In addition, we can also find that the approaches in group C achieve higher performance for multi-step prediction tasks (i.e., C1 v.s. A1 and C2 v.s. A2). These results also validate the robustness of the proposed framework for multi-step prediction by reducing the accumulated errors by BE representation.

Similarly, the BiLSTM architecture (C2) also suffers from larger prediction errors in the multi-step prediction due to the influence of the pseudo observations on the bidirectional sequence modeling, as explained in the last section.

2) Computational Performance Evaluation: The computation performance of each method is presented in Table III, and the experimental configurations are the same as that described in Section IV-B. In this evaluation, the batch size of all the experiments is set to 1024 in the training process, while it is set to 1 during the prediction phase. In the training phase, the AvgTC of proposed MLC-based methods (B1, C1, C2) is much more time-consuming than the deep regression-based methods (A1, A2). On the one hand, the BE representation transformation process (binary encoding) increases the time costs. On the other hand, the output quantization process of the MLC-based methods and its backpropagation of MSE loss also requires extra expenditure. In the prediction phase, the KF-based method (A3) achieves the best computational performance since it has fewer parameters than deep learning based methods. The MLC-Based methods are slightly inferior to the baselines due to more parameters and extra binary encoding/decoding operations. Note that the results of the AvgTC evaluation in the training process benefit from the parallel computing capability of GPUs. In practice, the time cost of training is larger than that of the prediction stage due to the backpropagation and parameter update of the network. Specifically, the MSE loss computation in the training process is a time costly operation that concerns the outputs quantization, transformation, and loss calculation in this work.

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

TABLE II
RESULTS OF THE EXPERIMENT GROUP C

Methods	Prediction steps	MAE ↓						RMSE ↓					
		Lon	Lat	Alt	Vx	Vy	Vz	Lon	Lat	Alt	Vx	Vy	Vz
C1 (LSTM / MBC / 0.001)	1	0.0051	0.0041	1.39	3.60	3.70	0.40	0.0667	0.0580	10.10	12.01	11.68	1.36
	3	0.0068	0.0054	2.40	5.42	5.69	0.57	0.0856	0.0763	12.59	16.50	16.98	1.76
	9	0.0131	0.0107	6.56	9.29	10.02	0.83	0.1091	0.0943	21.45	25.15	27.11	2.26
	15	0.0208	0.0181	11.54	12.16	13.26	0.95	0.1259	0.1150	32.50	30.48	33.23	2.45
C2 (BiLSTM / MBC / 0.001)	1	0.0037	0.0037	1.25	3.29	3.36	0.38	0.0570	0.0591	9.43	11.57	11.05	1.31
	3	0.0072	0.0071	2.52	5.20	5.45	0.55	0.0965	0.1069	14.17	16.56	16.89	1.73
	9	0.0188	0.0187	7.12	9.33	9.97	0.83	0.1667	0.1881	24.95	25.96	27.56	2.24
	15	0.0326	0.0319	12.17	12.44	13.35	0.95	0.2212	0.2445	35.76	31.85	33.95	2.43

TABLE III
COMPARISON OF COMPUTATIONAL PERFORMANCE AMONG THE PROPOSED METHOD AND BASELINES

Phase	AvgTC (ms)					
	A1	A2	A3	B1	C1	C2
Train	0.16	0.23	-	7.36	7.56	7.82
Prediction	5.47	8.18	0.04	12.39	10.21	15.72

It can be found from the AvgTC results that although the time costs of the proposed MLC-based methods are slightly higher than the baselines, they still meet the real-time predictions (millisecond-level overhead) for the TP tasks. Moreover, our model can also perform the inference process in a parallel manner by integrating the flight trajectories into a batch, which will further improve computational efficiency.

D. Visualization and Qualitative Analysis

In this section, we qualitatively analyze the proposed FlightBERT framework and baselines by visualizing the flight trajectories prediction results with some typical scenarios in real-time ATC procedure. A total of six ATC scenarios are visualized in Fig. 5, including climbing, climbing and turning right, descending, maintain, turn, climbing and maintain. Except for the "maintain" trajectory, other flight scenarios can be considered to have sufficient maneuverability, which also validates the robustness of the proposed framework in complex ATC environments.

As shown in Fig. 5, all trajectories are depicted in a 3D space, the initial inputs are the observed trajectory containing 9 points (corresponding to 3 minutes trajectory). The predicted trajectories are generated by the comparative approaches by multi-step prediction, i.e., with 15 steps (i.e., 5 minutes trajectory in the future).

It is clear that the proposed framework achieves the highest prediction performance based on the visualization results. In addition, the deep regression-based approaches (A1, A2) generate larger prediction errors with the increase of the prediction step. They can predict the future trajectory accurately in 1-3 prediction steps when the transition patterns can be clearly captured from the observation trajectory (such as Fig. 5b and Fig. 5d). However, with the increasing of prediction steps, the trajectories predicted by the baselines gradually deviate from the ground truth. The KF-based model achieved comparable performance with the proposed methods when the flight under the linear transition patterns (such as Fig. 5a and Fig. 5d). Similarly, it also failed to capture complex or non-linear transition patterns in the prediction process (such as Fig. 5c and Fig. 5e).

In contrast, the proposed framework not only captured the flight motion patterns in a few prediction steps but also can predict the movement trends in the future. As shown in Fig. 5e, the intent of the turn, which will take place 3 minutes later, is accurately predicted by the proposed framework, while other baselines incorrectly estimate the flight intent. On the one hand, the proposed FlightBERT framework can capture the transition patterns from the observations to output the high-confidence predictions, which further support the multi-step prediction process. On the other hand, we believe that the implicit knowledge, such as waypoint, route, is learned from the historical trajectory data in the training process, which is conducive to predicting the flight movement intents. It also proves that the proposed BE representation and trajectory embedding module are beneficial to mining the high-level trajectory features.

In addition, unreliable prediction results are even generated by baselines in the first prediction step, when the observation trajectory with complex transition patterns (Fig. 5c, descending). For high-maneuverability prediction, the baselines predict the trajectory in a wrong flight intent (Fig. 5c, 5e, and 5f), which is detrimental to the real-time ATC applications.

E. Ablation Study

In this section, to validate the effectiveness and contributions of the proposed ACoAtt block and hybrid constrained loss in the proposed FlightBERT framework, the following ablation experiments are designed:

- D: A total of three experiments are designed in this group, which are built by removing the ACoAtt block from the experiment B1, C1, and C2 architectures. In other words, the D1, D2, D3, are conducted on the proposed FlightBERT framework without ACoAtt block and using Transformer, LSTM, and BiLSTM backbone network.
- E: In this experiment, only the BCE loss is applied to optimize the proposed FlightBERT framework.

The results of mentioned experiments are listed in Table IV. In group D, by removing the ACoAtt block, the comparative approaches suffer from larger prediction errors in terms of the MAE and RMSE for all the attributes and prediction steps. The results can confirm the effectiveness of the proposed AcoAtt block, which can effectively and explicitly capture

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

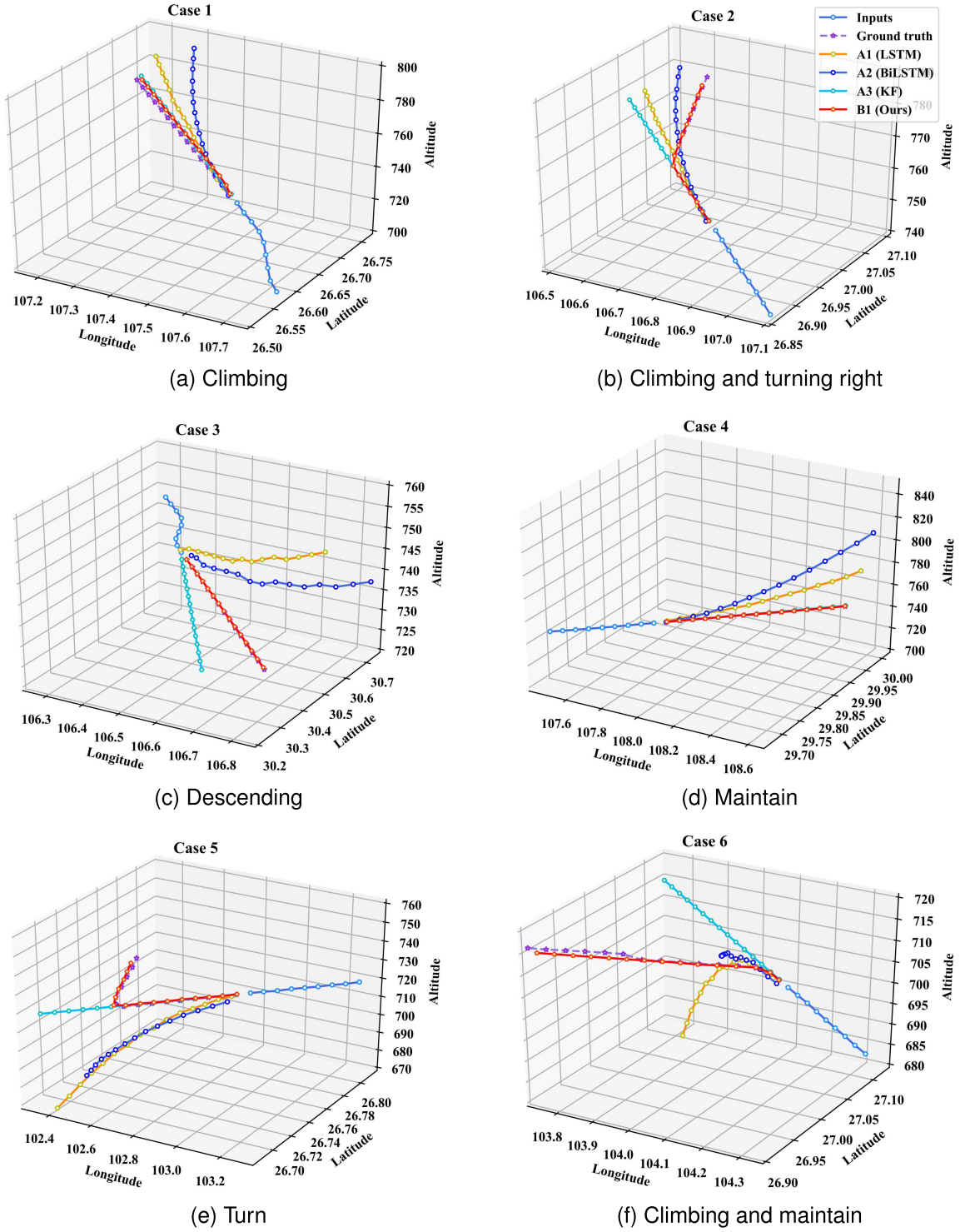


Fig. 5. Visualization of the trajectory prediction results in some ATC case with the experiment A1 (LSTM), A2 (BiLSTM), A3 (KF), B1 (The proposed), in which the altitude is measured in 10m.

the inherent correlations between specific attributes with BE representations, which further improves the performance of the proposed framework. Meanwhile, the measurements in group D are still better than those in group A, which also supports the BE representation and MBC mechanism. Fortunately, the ACoAtt block can further enhance the prediction performance by utilizing the prior knowledge.

As to experiment E, it obtained inferior performance compared to other MBC-based models. Based on our experiment, the proposed FlightBERT framework also works well only using the BCE loss function in the training phase, but fails to obtain a qualified performance in the test phase. We believe that the proposed HC loss has the ability to guide the model to learn the intra and inter correlations between the *n-hot* vector

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
RESULTS OF THE ABLATION STUDY

Methods	Prediction steps	MAE ↓						RMSE ↓					
		Lon	Lat	Alt	Vx	Vy	Vz	Lon	Lat	Alt	Vx	Vy	Vz
D1 (Transformer without ACoAtt)	1	0.0043	0.0042	1.58	3.45	3.65	0.40	0.0541	0.0589	12.86	11.51	11.81	1.34
	3	0.0068	0.0066	2.86	5.44	5.87	0.56	0.0822	0.0870	16.51	16.75	17.65	1.76
	9	0.0152	0.0145	7.48	9.82	10.73	0.85	0.1131	0.1151	25.54	26.41	28.62	2.28
	15	0.0259	0.0243	12.77	13.18	14.50	0.98	0.1503	0.1414	36.62	32.53	35.39	2.48
D2 (LSTM without ACoAtt)	1	0.0051	0.0047	1.47	3.64	3.63	0.39	0.0807	0.0745	12.02	12.45	11.72	1.33
	3	0.0076	0.0066	2.61	5.65	5.79	0.55	0.0991	0.0893	14.82	17.33	17.42	1.74
	9	0.0164	0.0139	6.97	9.91	10.40	0.83	0.1247	0.1065	23.35	26.45	27.89	2.24
	15	0.0278	0.0235	11.98	13.12	13.82	0.95	0.1612	0.1366	34.18	32.27	34.14	2.44
D2 (BiLSTM without ACoAtt)	1	0.0047	0.0040	1.43	3.37	3.40	0.38	0.0767	0.0640	11.45	11.80	11.05	1.30
	3	0.0085	0.0067	2.63	5.28	5.48	0.55	0.1269	0.0948	14.90	16.51	16.67	1.72
	9	0.0201	0.0161	7.13	9.50	10.03	0.82	0.1868	0.1374	24.54	26.02	27.18	2.23
	15	0.0333	0.0273	12.18	12.65	13.47	0.95	0.2305	0.1683	35.30	31.86	33.68	2.43
E (Only BCE Loss)	1	0.0055	0.0056	1.75	3.81	4.05	0.42	0.0723	0.0766	13.32	12.68	12.66	1.38
	3	0.0081	0.0081	3.21	5.89	6.40	0.59	0.0970	0.1008	17.63	17.75	18.65	1.81
	9	0.0170	0.0174	8.08	10.59	11.63	0.87	0.1230	0.1410	27.24	27.93	30.03	2.32
	15	0.0281	0.0282	13.47	14.14	15.64	0.99	0.1640	0.1748	38.44	34.24	36.97	2.50

of each attribute by combining the BCE and MSE loss, which further alleviates the high-dimensional prediction errors of the predicted *n-hot* vector. Note that we also attempt to train the proposed FlightBERT framework with the MSE loss, but the network even fails to reach a training converge.

In summary, the ablation study demonstrates that all the proposed ACoAtt block and HC loss are essential to improve the performance of the proposed FlightBERT framework.

F. Discussions

1) Error Analysis: From the abovementioned analysis, it can be concluded that the proposed FlightBERT framework (with BE representation and MBC mechanism) learns the desired knowledge from a large number of historical data in the training phase and can capture the transition patterns from the input trajectory in the inference process. On the one hand, the proposed BE representation transforms scalar values of trajectory attributes to the *n-hot* vectors to extract the high-level abstract features. On the other hand, the proposed model achieves better robustness than baselines and is not sensitive to the prediction errors of pseudo observation.

As to the larger RMSE for predicting the point at the next time step, based on our experimental analysis, we found that the primary contributor of the errors is the prediction errors on higher bits. The misclassification of higher bits will cause heavy burdens on the final performance. *For instance*, if repeat the BE "0110 0100" (decimal 100) with minibatch size 10, and all the predictions are incorrectly recognized as "0100 0100" (decimal 68), the BCE loss only accounts for one binary, while the MAE is 32, and the RMSE will up to 101.2.

Although the HC loss is proposed to alleviate the misclassification on higher bits, it can not completely overcome such problems due to the MBC-based architecture. In practice, this issue can be further tackled from the engineering perspective. One practical solution is to set a pre-defined sector ROI to control the model outputs, depending on the expert prior based on current observation for the model inference in the real-time TP tasks. Once the model outputs fail to fall within the preset ROI, it will be considered unreliable prediction results and stop the multi-step prediction process.

To further evaluate the effects of this problem for the TP tasks, the MAE of each flight trajectory on the test set is evaluated using model B1 with the $1^{st}, 3^{rd}, 9^{th}$, and 15th, prediction steps. The MAE of six attributes is presented by the boxplots in Fig. 6. Note that the outliers (about 3%) are not presented in the figures to improve the readability of the boxplots. It can be found that the error is accumulating as the prediction steps increases and fluctuate within a normal range for all the six attributes.

Overall, although there are some outliers in the prediction results of the proposed BE representation, the proposed framework still shows great improvements over baselines in the TP task, as well as the multi-step prediction.

2) Evaluation by Different Flight Phases: To evaluate the robustness and performance of the proposed method in the different flight phases, the trajectories of the test set are further split into three phases, i.e., perform predictions on climbing, en-route, and descending/approaching phase, respectively. In the experiments, the proposed approach with the model B1 configurations is used to perform the predictions with 1, 3, 9, and 15 steps, which corresponds to 20 seconds and 1 minute, 3 minutes, and 5 minutes, respectively.

The results are illustrated in Table V. The performance in the en-route phase is better than in the climbing and descending phase. It can be attributed that the transition patterns of trajectories in the en-route phase are approximate to linear and can be captured easily to predict the future trajectory. In contrast, more complex maneuvering is always performed in the climbing and descending/approaching phases, which is challenging to learn its transition patterns.

In general, the MAE and RMSE metrics of the longitude and latitude in three flight phases fluctuate with small gaps. It can be also learned from the results that the V_z is the driven status between en-route and climbing/descending phases, i.e., the prediction errors of V_z and altitude are mainly made from the climbing and descending phase. Obviously, the en-route phase has few altitude changes whereas the maneuvering is

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

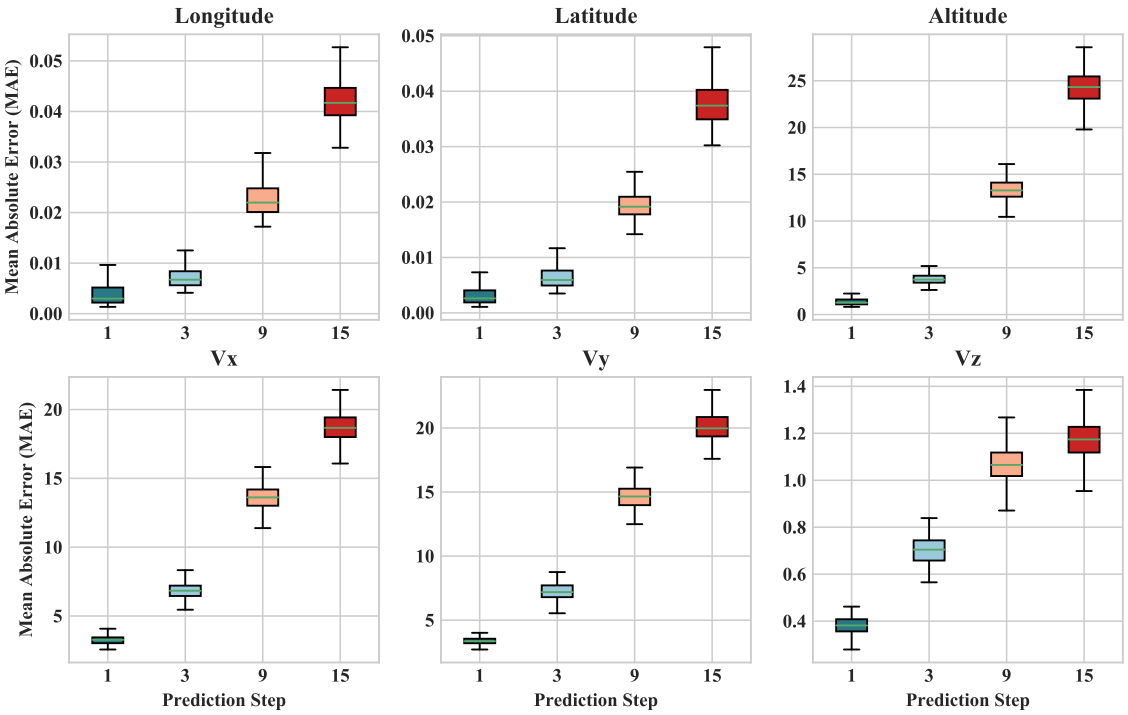


Fig. 6. Boxplots of six attributes for each prediction step.

TABLE V
PERFORMANCE OF MODEL B1 EVALUATES IN DIFFERENT FLIGHT PHASES

Phase	Prediction steps	MAE ↓						RMSE ↓					
		Lon	Lat	Alt	Vx	Vy	Vz	Lon	Lat	Alt	Vx	Vy	Vz
Climbing	1	0.0046	0.0040	7.46	4.74	5.33	2.04	0.0521	0.0512	28.08	12.64	14.68	3.09
	3	0.0081	0.0068	13.39	8.30	9.99	2.54	0.0202	0.0188	29.34	17.58	21.01	3.27
	9	0.0202	0.0188	29.34	17.58	21.00	3.27	0.1108	0.0828	53.88	36.82	46.25	4.41
	15	0.0361	0.0334	46.38	23.55	27.15	3.61	0.1426	0.0988	75.73	44.73	55.77	4.69
En-route	1	0.0042	0.0033	0.09	2.79	2.82	0.0002	0.0612	0.0458	1.90	10.61	9.67	0.0078
	3	0.0056	0.0049	0.11	4.10	4.19	0.0003	0.0735	0.0700	2.31	14.26	13.80	0.0151
	9	0.0116	0.0098	0.19	6.74	7.01	0.0005	0.0952	0.0862	4.09	21.18	21.36	0.0231
	15	0.0185	0.0156	0.23	8.73	9.04	0.0007	0.1151	0.1043	5.38	25.47	25.73	0.0302
Descending / Approaching	1	0.0041	0.0047	3.94	4.04	3.77	1.02	0.0504	0.0726	18.85	12.40	10.75	1.95
	3	0.0055	0.0049	6.50	6.99	6.58	1.27	0.0542	0.0477	21.72	19.50	17.56	2.29
	9	0.0173	0.0126	16.38	14.97	13.28	1.81	0.1114	0.0565	34.02	35.92	30.80	2.89
	15	0.0271	0.0222	25.42	20.21	17.66	2.11	0.1055	0.0725	46.10	43.15	36.78	3.17

more frequent in the vertical direction of the climbing and descending phase. The experiment results proved that our model can be made predictions well in all the flight phases, which also has been visualized and discussed in Section IV-D.

V. CONCLUSION AND FUTURE WORK

In this paper, we present a novel framework, called FlightBERT, to tackle the flight TP task. The BE representation is proposed to enhance the input capability and eliminate the influence of the normalization algorithm. Compared with traditional regression based approaches, the proposed FlightBERT framework formulates the TP task as a multi binary classification problem by the proposed BE trajectory representation. The trajectory embedding module is designed to project the trajectory attributes to the embedding space by considering the correlations between specific attributes. A combined loss function is designed to optimize the proposed framework. Experimental results on a large-scale real-world dataset demonstrate that the proposed framework outperforms the baselines by both the quantitative and qualitative evaluations. All technical improvements are confirmed by the ablation study. The framework can be generalized to other backbone networks, which confirms the robustness of the BE representation and MBC mechanism. In addition, the proposed framework is able to achieve higher performance for multi-step prediction for different backbones, which improves the practicability of the proposed framework in real-time ATC applications.

In the future, we will focus on exploring the solutions to reduce the outliers (misclassification in high BE bits) of the prediction trajectory in the FlightBERT framework which further improves its robustness. In addition, generalizing the BE representation to the other ATC applications and trajectory prediction tasks is also our interest.

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] Y. Lin, J.-W. Zhang, and H. Liu, "Deep learning based short-term air traffic flow prediction considering temporal–spatial correlation," *Aerosp. Sci. Technol.*, vol. 93, Oct. 2019, Art. no. 105113.

[2] W. Liu and I. Hwang, "Probabilistic trajectory prediction and conflict detection for air traffic control," *J. Guid., Control, Dyn.*, vol. 34, no. 6, pp. 1779–1789, 2011.

[3] Z. Chen, D. Guo, and Y. Lin, "A deep Gaussian process-based flight trajectory prediction approach and its application on conflict detection," *Algorithms*, vol. 13, no. 11, p. 293, Nov. 2020.

[4] Z. Wang, M. Liang, and D. Delahaye, "A hybrid machine learning model for short-term estimated time of arrival prediction in terminal manoeuvring area," *Transp. Res. C, Emerg. Technol.*, vol. 95, pp. 280–294, Oct. 2018.

[5] "Common trajectory prediction structure and terminology in support of SESAR & NextGen," FAA/EUROCONTROL AP 16, Brussels, Belgium, Tech. Rep., 2010.

[6] Z. Shi, M. Xu, and Q. Pan, "4-D flight trajectory prediction with constrained LSTM network," *IEEE Trans. Intell. Transp.*, vol. 22, no. 11, pp. 7242–7255, Nov. 2021.

[7] Y. Lin, L. Deng, Z. Chen, X. Wu, J. Zhang, and B. Yang, "A real-time ATC safety monitoring framework using a deep learning approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 11, pp. 4572–4581, Nov. 2020.

[8] X. M. Tang, L. Zhou, Z. Y. Shen, and M. Tang, "4D trajectory prediction of aircraft taxiing based on fitting velocity profile," *Aeronautical Computing Technique*, vol. 45, no. 1, pp. 1–12, 2015.

[9] M. Daigle et al., "Real-time prediction of safety margins in the national airspace," in *Proc. 17th AIAA Aviation Technol., Integr., Oper. Conf.*, 2017, p. 4388.

[10] H. Yan, G. Huang, H. Wang, and R. Shu, "Application of unscented Kalman filter for flying target tracking," in *Proc. Int. Conf. Inf. Sci. Cloud Comput.*, Dec. 2013, pp. 61–66.

[11] D. P. Thipphavong, C. A. Schultz, A. G. Lee, and S. H. Chan, "Adaptive algorithm to improve trajectory prediction accuracy of climbing aircraft," *J. Guid., Control, Dyn.*, vol. 36, no. 1, pp. 15–24, Jan. 2013.

[12] Z.-J. Wu, S. Tian, and L. Ma, "A 4D trajectory prediction model based on the BP neural network," *J. Intell. Syst.*, vol. 29, no. 1, pp. 1545–1557, Aug. 2020.

[13] W. Zeng, Z. Quan, Z. Zhao, C. Xie, and X. Lu, "A deep learning approach for aircraft trajectory prediction in terminal airspace," *IEEE Access*, vol. 8, pp. 151250–151266, 2020.

[14] Z. Shi, M. Xu, Q. Pan, B. Yan, and H. Zhang, "LSTM-based flight trajectory prediction," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Rio de Janeiro, Brazil, Jul. 2018, pp. 1–8.

[15] A. Nuic, "User manual for the base of aircraft data (BADA) revision 3.10," *Atmosphere*, vol. 2010, p. 1–109, Feb. 2010.

[16] E. Gallo, J. Lopez-Leones, M. A. Vilaplana, F. A. Navarro, and A. Nuic, "Trajectory computation infrastructure based on BADA aircraft performance model," in *Proc. IEEE/AIAA 26th Digit. Avionics Syst. Conf.*, Oct. 2007, p. 1–13.

[17] M. Porretta, M.-D. Dupuy, W. Schuster, A. Majumdar, and W. Ochieng, "Performance evaluation of a novel 4D trajectory prediction model for civil aircraft," *J. Navigat.*, vol. 61, no. 3, pp. 393–420, Jul. 2008.

[18] J. Sun, J. Ellerbroek, and J. Hoekstra, "Modeling and inferring aircraft takeoff mass from runway ADS-B data," in *Proc. 7th Int. Conf. Res. Air Transp.*, 2016, pp. 1–5.

[19] L. Xi, Z. Jun, Z. Yanbo, and L. Wei, "Simulation study of algorithms for aircraft trajectory prediction based on ADS-B technology," in *Proc. Asia Simul. Conf. 7th Int. Conf. Syst. Simul. Sci. Comput.*, Oct. 2008, pp. 322–327.

[20] X. R. Li and V. P. Jilkov, "Survey of maneuvering target tracking. Part V: Multiple-model methods," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 41, no. 4, pp. 1255–1321, Oct. 2005.

[21] Y. Song, P. Cheng, and C. Mu, "An improved trajectory prediction algorithm based on trajectory data mining for air traffic management," in *Proc. IEEE Int. Conf. Inf. Autom.*, Jun. 2012, pp. 981–986.

[22] H. Jeung, H. T. Shen, and X. Zhou, "Mining trajectory patterns using hidden Markov models," in *Proc. Int. Conf. Data Warehousing Knowl. Discovery*. Cham, Switzerland: Springer, 2007, pp. 470–480.

[23] J. C. Nascimento, M. A. T. Figueiredo, and J. S. Marques, "Trajectory classification using switched dynamical hidden Markov models," *IEEE Trans. Image Process.*, vol. 19, no. 5, pp. 1338–1348, May 2010.

[24] J. Wiest, M. Höffken, U. Kreßel, and K. Dietmayer, "Probabilistic trajectory prediction with Gaussian mixture models," in *Proc. IEEE Intell. Vehicles Symp.*, Jun. 2012, pp. 141–146.

[25] H. Yan, B. Yang, and H. Yang, "Probabilistic approach to conformance monitoring using Gaussian processes," *AIAA J. Guid., Control, Dyn.*, vol. 40, no. 6, pp. 1403–1414, Jun. 2017.

[26] X.-R. Jiang, X.-X. Wen, M.-G. Wu, Z.-K. Wang, and X. Qiu, "A SVM approach of aircraft conflict detection in free flight," *J. Adv. Transp.*, vol. 2018, pp. 1–9, Dec. 2018.

[27] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 2, pp. 84–90, Jun. 2017.

[28] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. 9th Int. Conf. Learn. Represent. (ICLR)*, May 2021, pp. 1–11.

[29] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 1–16.

[30] Y. Lin, D. Guo, J. Zhang, Z. Chen, and B. Yang, "A unified framework for multilingual speech recognition in air traffic control systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 8, pp. 3608–3620, Aug. 2021.

[31] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Proc. Annu. Conf. Neural Inf. Process. Syst. (NeurIPS)*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds. New York, NY, USA: Curran, 2020, pp. 12449–12460.

[32] Z. Xu, W. Zeng, X. Chu, and P. Cao, "Multi-aircraft trajectory collaborative prediction based on social long short-term memory network," *Aerospace*, vol. 8, no. 4, p. 115, Apr. 2021.

[33] Z. Wang, M. Liang, and D. Delahaye, "Short-term 4D trajectory prediction using machine learning methods," in *Proc. 7th SESAR Innov. Days*, Belgrade, Serbia, Nov. 2017, pp. 1–9.

[34] X. Zhang and S. Mahadevan, "Bayesian neural networks for flight trajectory prediction and safety assessment," *Decis. Support Syst.*, vol. 131, Apr. 2020, Art. no. 113246.

[35] Y. Pang, X. Zhao, H. Yan, and Y. Liu, "Data-driven trajectory prediction with weather uncertainties: A Bayesian deep learning approach," *Transp. Res. C, Emerg. Technol.*, vol. 130, Sep. 2021, Art. no. 103326.

[36] L. Ma and S. Tian, "A hybrid CNN-LSTM model for aircraft 4D trajectory prediction," *IEEE Access*, vol. 8, pp. 134668–134680, 2020.

[37] H. Sahfienya and A. C. Regan, "4D flight trajectory prediction using a hybrid deep learning prediction method based on ads-b technology: A case study of hartsfield–jackson Atlanta international airport(ATL)," 2021, *arXiv:2110.07774*.

[38] Y. Liu and M. Hansen, "Predicting aircraft trajectories: A deep generative convolutional recurrent neural networks approach," 2018, *arXiv:1812.11670*.

[39] H. Georgiou, N. Pelekis, S. Sideridis, D. Scarlatti, and Y. Theodoridis, "Semantic-aware aircraft trajectory prediction using flight plans," *Int. J. Data Sci. Anal.*, vol. 9, no. 2, pp. 215–228, Mar. 2020.

[40] P. Liu, X. Qiu, X. Chen, S. Wu, and X. Huang, "Multi-timescale long short-term memory neural network for modelling sentences and documents," in *Proc. Conf. Empirical Methods Natural Lang. Process., (EMNLP)*, L. Màrquez, C. Callison-Burch, J. Su, D. Pighin, and Y. Marton, Eds. Lisbon, Portugal: The Association for Computational Linguistics, Sep. 2015, pp. 2326–2335.

[41] Y. Lin et al., "ATCSpeechNet: A multilingual end-to-end speech recognition framework for air traffic control systems," *Appl. Soft Comput.*, vol. 112, Nov. 2021, Art. no. 107847.

[42] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.

[43] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst., Annu. Conf. Neural Inf. Process. Syst.*, Long Beach, CA, USA, Dec. 2017, pp. 5998–6008.

[44] D. Guo, J. Zhang, B. Yang, and Y. Lin, "A comparative study of speaker role identification in air traffic communication using deep learning approaches," 2021, *arXiv:2111.02041*.

[45] M.-L. Zhang and Z.-H. Zhou, "A review on multi-label learning algorithms," *IEEE Trans. Knowl. Data Eng.*, vol. 26, no. 8, pp. 1819–1837, Jun. 2014.

[46] A. Radford, K. Narasimhan, T. Salimans, and I. Sutskever, "Improving language understanding by generative pre-training," Openai.com. Accessed: Nov. 6, 2022. [Online]. Available: https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf

[47] Q. Wen et al., "Transformers in time series: A survey," 2022, *arXiv:2202.07125*.

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.

[48] R. E. Kalman, "A new approach to linear filtering and prediction problems," *J. Basic Eng.*, vol. 82, no. 1, pp. 35–45, Mar. 1960.
[49] Y. Bengio, N. Léonard, and A. Courville, "Estimating or propagating gradients through stochastic neurons for conditional computation," 2013, *arXiv:1308.3432*.



Dongyue Guo received the M.E. degree from Sichuan University, Chengdu, China, in 2020, where he is currently pursuing the Ph.D. degree with the National Key Laboratory of Fundamental Science on Synthetic Vision, College of Computer Science.

His research interests include automatic speech recognition and flight trajectory prediction.



Edmond Q. Wu (Senior Member, IEEE) received the Ph.D. degree in controlling theory and engineering from Southeast University, Nanjing, China, in 2009.

He is currently a Professor with the Key Laboratory of System Control and Information Processing, Ministry of Education, Shanghai Jiao Tong University, China. He is also with the Science and Technology on Avionics Integration Laboratory, China National Aeronautical Radio Electronics Research Institute, Shanghai, China. His research interests include deep learning, fatigue recognition, and human–machine interaction. He is currently an Associate Editor of IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE TRANSACTIONS ON INTELLIGENT VEHICLES, and IEEE TRANSACTIONS ON COGNITIVE AND DEVELOPMENTAL SYSTEMS.



Yuankai Wu (Member, IEEE) received the Ph.D. degree from the Beijing Institute of Technology, Beijing, China, in 2019.

He was a Visiting Ph.D. student at the University of Wisconsin, Madison, WI, USA. He is a Postdoctoral Researcher with McGill University, Montreal, QC, Canada, supported by the Institute for Data Valorization (IVADO). His research interests include intelligent transportation systems, intelligent energy management, and machine learning.



Jianwei Zhang received the Ph.D. degree from Sichuan University, Chengdu, China, in 2008.

He has been taught and conducted research at Sichuan University since 1993. He has published more than 30 articles. His research interests include air traffic management and intelligent image analysis and processing. He received the National Science and Technology Progress Award in China.



Rob Law is the University of Macau Development Foundation (UMDF) Chair Professor of Smart Tourism. He is also an Honorary Professor of several other reputable universities. Prior to joining the University of Macau in July 2021, he has worked in industry organizations and academic institutes. He is an Active Researcher. He has received more than 90 research related awards and accolades (e.g. recognized as the most prolific tourism/hospitality researcher in the world over two decades from 2000 to 2019—International Journal of Contemporary Hospitality Management, 2021), as well as millions of USD external and internal research grants. He has edited four books and published more than 1,000 research papers (including hundreds of articles in first-tier academic journals). His publications have received more than 49,800 citations, with H-index/i 10-index = 103/454 (http://scholar.google.com.hk/, accessed on February 14, 2022). In addition, he serves different roles for more than 200 research journals and the chair/committee member of more than 180 international conferences.



Yi Lin (Member, IEEE) received the Ph.D. degree from Sichuan University, Chengdu, China, in 2019.

He currently works as an Associate Professor with the College of Computer Science, Sichuan University. He is also a Visiting Scholar with the Department of Civil and Environmental Engineering, University of Wisconsin–Madison, Madison, WI, USA. His research interests include air traffic flow management and planning, machine learning, and deep-learning-based air traffic management applications.

Authorized licensed use limited to: UNIVERSITA BICOCCA MILANO. Downloaded on September 25,2026 at 20:30:25 UTC from IEEE Xplore. Restrictions apply.